Unit 9, Lesson 6: Rewriting Equivalent Numerical Expressions

Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Learning Target

☐ I can determine if two numerical expressions are equivalent by rewriting each in the same form.

9.6.1 Warm-Up: Two Truths and a Lie

1. Students were asked to write two truths and a lie, or two true statements and one false statement, about the associative property of addition. One student's work is shown.

Associative Property of Addition

A. $(5+4)+3 \quad 5+(4+3)$

B. $3+(4+5) = (3+4)+5$

C. $(4+3)+5 = 4+(3+5)$

Which statement is the lie? Justify your response.

9.6.2 Guided Instruction: Rewriting Equivalent Expressions

1. Consider the expressions $18 + 12(30 + 7)$ and $18 + 12(7 + 30)$.

   a. Describe what is different about the expressions.

   b. Evaluate $18 + 12(30 + 7)$ and $18 + 12(7 + 30)$.

$18 + 12(30 + 7)$	$18 + 12(7 + 30)$

   c. Are the two expressions equivalent?

   d. Was it necessary to evaluate the two expressions to determine whether they were equivalent or not?

2. The
   - ○ associative
   - ○ commutative
   - ○ distributive

   property of addition states that the order in which two numbers are added does not matter.

3. For what other mathematical operations does this property hold true?

4. Eric thinks the expressions shown are equivalent to each other.

$18 + 12(30 + 7)$ $\qquad$ $18 + 360 + 84$ $\qquad$ $30(30 + 7)$

Without evaluating each expression, explain if he is correct.

5. The
- ○ associative
- ○ commutative
- ○ distributive

property of operations can be used to show that $18 + 12(30 + 7) = 18 + 360 + 84$.

Equivalent expressions can be rewritten in different forms using the properties of operations.

The following properties of operations apply to all rational numbers.

The ***distributive property of multiplication over addition or subtraction*** states that a factor is "distributed" to all terms within a set of parentheses.

The ***commutative property*** states that for addition and multiplication, the order in which the operation is performed does not change the value of the expression.

The ***associative property*** states that for addition and multiplication, the way in which numbers are grouped does not change the value of the expression.

6. Work with your partner to rewrite the expression $5 - 6.2$ using the commutative property.

7. Work with your partner to rewrite the following expression in **two** equivalent forms using the properties of operations.

$$\left(\frac{2}{5} + 4\right) + 6 - \frac{1}{2}(5 - 6.2)$$

9.6.3 Try It: Rewriting Equivalent Expressions

1. Maria was asked to rewrite the following expression in a variety of ways.

$$5\frac{1}{4} - 4(-3.5 + 7) + \left(\frac{3}{5} + 5\right) + 8.25$$

She came up with four different expressions, but she made some errors. Without evaluating, decide which expressions are equivalent using properties of operations. Then, identify the errors she made.

Expression	Is it equivalent? Justify why or why not.
$5\frac{1}{4} - 14 - 28 + \left(\frac{3}{5} + 5\right) + 8.25$	
$5\frac{1}{4} - 4(-3.5 + 7) + \frac{3}{5} + (8.25 + 5)$	
$5\frac{1}{4} + 14 - 28 + \frac{3}{5} + (5 + 8.25)$	
$1.25(-3.5 + 7) + \left(5 + \frac{3}{5}\right) + 8.25$	

9.6.4 Your Turn!

1. Rewrite the following expression in **two** ways using the properties of operations.

$$\frac{12}{35} \cdot \frac{1}{2} + \frac{1}{5} + \left(\frac{2}{5} + \frac{3}{7}\right)$$

9.6.5 Wrap-Up: Error Analysis

1. Juan was asked to rewrite the expression

   $\left(-6 + 2\frac{1}{2} + 3\right) + 4 - 2\left(-3\frac{2}{3}\right)$ using the properties of operations.

   Juan wrote $-6 + \left(3 + 2\frac{1}{2} + 4\right) + 7\frac{1}{3}$.

   Determine if Juan is correct. Show your work or explain your thinking.